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Model Zoos: A Dataset of Diverse Populations of Neural Network Models

Konstantin Schürholt[1], **Diyar Taskiran**[1], **Boris Knyazev**[2], **Xavier Giró-i-Nieto**[3], **Damian Borth**[1]
[1] AIML Lab, School of Computer Science, University of St.Gallen
[2] Samsung - SAIT AI Lab, Montreal
[3] Institut de Robòtica i Informàtica Industrial, Universitat Politècnica de Catalunya

Abstract

In the last years, neural networks (NN) have evolved from laboratory environments to the state-of-the-art for many real-world problems. It was shown that NN models (i.e., their weights and biases) evolve on unique trajectories in weight space during training. Following, a population of such neural network models (referred to as *model zoo*) would form structures in weight space. We think that the geometry, curvature and smoothness of these structures contain information about the state of training and can reveal latent properties of individual models. With such *model zoos*, one could investigate novel approaches for (i) model analysis, (ii) discover unknown learning dynamics, (iii) learn rich representations of such populations, or (iv) exploit the *model zoos* for generative modelling of NN weights and biases. Unfortunately, the lack of standardized *model zoos* and available benchmarks significantly increases the friction for further research about populations of NNs. With this work, we publish a novel dataset of *model zoos* containing systematically generated and diverse populations of NN models for further research. In total the proposed model zoo dataset is based on eight image datasets, consists of 27 *model zoos* trained with varying hyperparameter combinations and includes 50'360 unique NN models as well as their sparsified twins, resulting in over 3'844'360 collected model states. Additionally, to the model zoo data we provide an in-depth analysis of the zoos and provide benchmarks for multiple downstream tasks. The dataset can be found at www.modelzoos.cc.

1 Introduction

The success of Neural Networks (NN) is surprising, considering the hard optimization problem to be solved during training of NNs. Specifically, NN training is NP-complete [2], the loss surface and optimization problem are non-convex [9, 17, 31] and the parameter space to fit during training is high dimensional [3]. Additionally, NN training is sensitive to random initialization and hyperparameter selection [19, 32]. Together, this leads to an interesting characteristic of NN training: given a dataset and an architecture, different random initializations or hyperparameters lead to different minima on the loss surface and therefore result in different model parameters (i.e., weights and biases). Consequently, multiple training results in different NN models. The resulting population of NN (referred to as *model zoo*) is an interesting object to study: Do individual models of the model zoo have something in common? Do they form structures in weight space? What can we infer from such structures? Can we learn representations of them? Lastly, can such structures be exploited to generate new models with controllable properties?

These questions have been partially answered in prior work. Theoretical and empirical work demonstrates increasingly well-behaved loss surfaces for growing number of parameters [17, 8, 32]. The shape of the loss surface and the starting point is determined by hyperparameters and the initialization,

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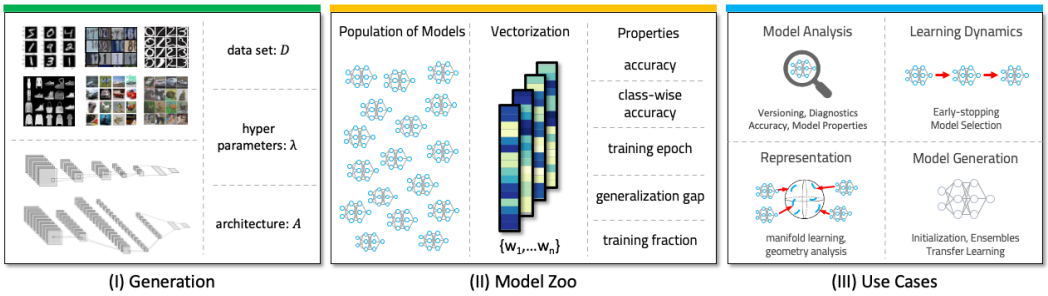

(I) Generation (II) Model Zoo (III) Use Cases

Figure 1: The proposed dataset of model zoos is trained on several image dataset with two CNN architectures and a multiple configurations of hyperparameters. The resulting population of neural network models is vectorized and made available with all meta-data such as the generating factors of the model zoo as well as the model properties such as accuracy, generalization gap and other. Potential use cases are (a) model property prediction, (b) inference of learning dynamics, (c) representation learning, or (d) model generation.

respectively [32]. NN training navigates the loss surface with iterative, gradient-based update schemes smoothed by momentum. The step length along a trajectory as well as the curvature are determined by the change of the loss as well as how aligned the subsequent updates are [4, 49]. Together, these findings suggest that populations of NN models evolve on unique and smooth trajectories in weight space. Related work has empirically confirmed the existence of such structures in NNs [11], demonstrated the feasibility to learn representations of them, showed that they encode information on model properties [55, 12, 50] and can be used to generate unseen models with desirable properties [52, 51, 64, 26] To thoroughly answer the questions above, a large and systematically created dataset of model weights is necessary.

Unfortunately, so far only few model zoos with specific properties have been published [55, 12, 54, 50]. While many machine learning domains have standardized datasets, there is no model zoo nor a benchmark to evaluate and compare against. The lack of a standardized model zoos has three significant disadvantages: (i), existing model zoos are usually designed for a specific purpose and of limited general utility. Their design space is rather sparse, covering only small portions of all available hyperparameter combinations. Moreover, some existing zoos are generated on synthetic tasks and are small, containing only a small population of models; (ii), researchers have to choose between using an existing zoo or generating a new one for each new experiment, weighing disadvantages of existing zoos against the effort and computational resources required to generate a new zoo; (iii), a new model zoo causes subsequent work to lose comparability to existing research. Therefore, the lack of a benchmark model zoo significantly increases the friction for new research.

Our contributions: To study the behaviour of populations of NNs, we publish a large-scale model zoo of diverse populations of neural network models with controlled generating factors of model training. Special care has been taken in their design and the used protocols for training. To do so, we have defined and restricted the generating factors of model zoo training to achieve desired zoo characteristics.

The zoos are trained on eight standard image classification datasets, with a broad range of hyperparameters and contain thousands of configurations. Further, we add sparsified *model zoo twins* to each of these zoos. All together, the zoos include a total of 50'360 unique image classification NNs, resulting in over 3'844'360 collected model states.

Potential use-cases for the model zoo include (a) model analysis for reliability, bias, fairness, or adversarial vulnerability, (b) inference of learning dynamics for efficiency gain, model selection or early stopping, (c) representation learning of such populations, or (d) model generation. Additionally, we present an analysis of the model zoos and a set of experimental setups for benchmarks on these use-cases and initial results as foundation for evaluation and comparison.

With this work we provide a standardized dataset of diverse model zoos connected to popular image datasets, its corresponding meta-data and performance evaluations to the machine learning research community. All data is made publicly available to foster community building around the topic and to provide a ground for use beyond the defined benchmark tasks. An overview of the proposed dataset and benchmark as well as potential use-cases can be found in Fig. 1

2 Existing Populations of Neural Networks Models

With the increase in usage of neural networks, requirements for evaluation, testing and certification have grown. Methods to analyze NN models may attempt to visualize salient features for a given class [62, 25, 61], investigate the robustness of models to specific types of noise [66, 7], predict model properties from model features [58, 24, 6] or compare models based on their activations [46, 42, 44] However, while most of these methods rely on common (image) datasets to train and evaluate their models, there is no common dataset of neural network models to compare the evaluation methods on. Model zoos as common evaluation datasets can be a step up to evaluate the evaluation methods.

There are only few publications who use model zoos. In [35], zoos of pre-trained models are used as teacher models to train a target model. Similarly, [53] propose a method to learn a combination of the weights of models from a zoo for a new task. [65] uses a zoo of GAN models trained with different methods to accelerate GAN training. To facilitate continual learning, [47] propose to generate zoos of models trained on different tasks or experiences, and to ensemble them for future tasks.

Larger model zoos containing a few thousand models are used in [55] to predict the accuracy of the models from their weights. Similarly, [12] use zoos of larger models to predict hyperparameters from the weights. In [16], a large collection of 3x3 convolutional filters trained on different datasets is presented and analysed. Other work identifies structures in the form of subspaces with beneficial properties [37, 56, 1]. [50] use zoos to learn self-supervised representations on the weights of the models in the zoo. The authors demonstrate that the learned representations have high predictive capabilities for model properties such as accuracy, generalization gap, epoch and various hyperparameters. Further, they investigate the impact of the generating factors of model zoos on their properties. [52, 51] demonstrate that learned representations can be instantiated in new models, as initialization for fine-tuning or transfer learning. This work systematically extends their zoos to more datasets and architectures.

3 Model Zoo Generation

The proposed model zoo datasets contain systematically generated and diverse populations of neural networks. Since the applicability of the model zoos for downstream tasks largely depends on the composition and properties of the zoos, special care has to be taken in their design and the used protocol for training. The entire procedure can be considered as defining and restricting the generating factors of model zoo training with respect to their latent relation of desired zoo characteristics. The described procedure and protocol could be also used as general blueprint for the generation of model zoos.

In our paper, the term architecture means the structure of a NN, i.e., set of operations and their connectivity. We use 'model' to denote an instantiating of an architecture with weights over all stages of training, 'model state' to denote the model with the specific state of weights at a specific training epoch, and the weights $\mathbf{w}$ to denote all trainable parameters (weights and biases).

3.1 Model Zoo Design

Generating Factors Following [55], we define the tuple $\{\mathcal{D}, \lambda, \mathcal{A}\}$ as a configuration of a model zoo's generating factors. We denote the dataset of image samples with their corresponding labels as $\mathcal{D}$. The NN architecture is denoted by $\mathcal{A}$. We denote the set of hyperparameters used for training, (*e.g.*, loss function, optimizer, learning rate, weight initialization, seed, batch-size, epochs) as λ. While dataset $\mathcal{D}$ and architecture $\mathcal{A}$ are fixed for a model zoo, λ provides not only the set of hyperparameters but also configures the ranges for individual hyperparameter such as learning rate for model zoo generation. Training with such differing configurations $\{\mathcal{D}, \lambda, \mathcal{A}\}$ results in a population of NN models i.e., the model zoo. We convert the weights and biases of each model to a vectorized form. In the resulting model zoo $\mathcal{W} = \{\mathbf{w}_1, ..., \mathbf{w}_M\}$, $\mathbf{w}_i$ denotes the flattened vector of the weights and biases of one trained NN model from the set of M models of the zoo.

Configurations & Diversity The model zoos have to be representative of real world models, but also diverse and span an interesting range of properties. The definition of diversity of model zoos, as well as the choice of how much diversity to include, is as difficult as in image datasets, e.g. [10, 13]. Model zoos can be diverse in their properties (i.e., performance) as well as in their generating factors λ, or in their weights $\mathbf{w}$. We aim at generating model zoos with a rich set of models and diversity in these aspects. As these zoo properties are effects of the generating factors, we tune the diversity of the generating factors and evaluate the diversity in Section 4.